UNIT III

Chapter 5

Digestion and Absorption

Chapter Outline

5.1 Digestive system

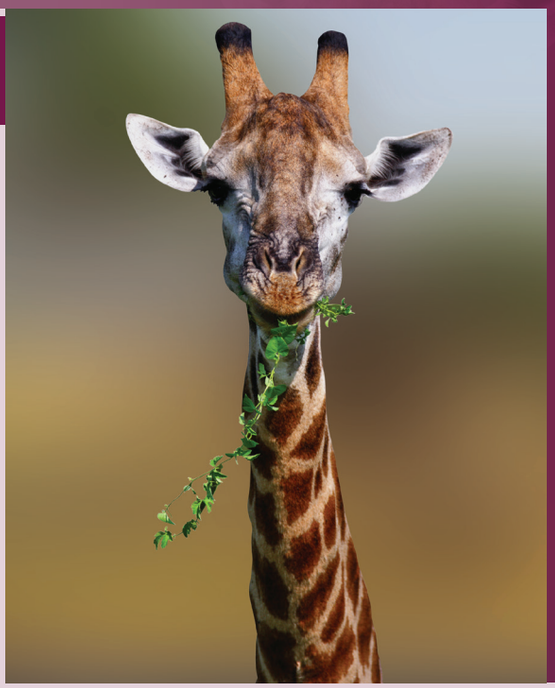
5.2 Digestion of food and role of digestive enzymes

5.3 Absorption and assimilation of proteins, carbohydrates and fats

5.4 Egestion

5.5 Caloric value of carbohydrates, proteins and fats

5.6 Nutritional and digestive disorders



Obtaining and utilizing nutrients is a fundamental process in all living organisms.

Learning Objectives:

- *Identifies and explains the major parts of the alimentary canal and digestive glands.*
- *Learns the process of digestion in various parts of the alimentary canal.*
- *Understands the role of enzymes in the process of digestion.*
- *Learns the symptoms of digestive disorders.*
- *Learns the role of nutrients in energy production, body building and maintenance and regulation of body activities.*
- *Creates awareness about the nutritional disorders and alimentary canal disorders.*

We all eat food. If you do not take breakfast in the morning how do you feel by noon? The food we eat provides energy and organic substances for growth and for replacement of worn and damaged tissues. It also regulates and coordinates the various activities that take place in our body. The components of our food are carbohydrates, proteins, lipids, vitamins, minerals, fibre and water. We obtain food from plant and animal sources. The food, we eat are macromolecules and cannot directly enter into our cells. These have to be broken into smaller micromolecules in absorbable forms, for which we need a digestive system. Plants however are autotrophs and synthesize their food, hence they do not require a digestive system. The primary function of the digestive system in the animals is to bring

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